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RESEARCH ARTICLE

Designing and Evaluating Online Health Consultation Interfaces: A Perspective of Physician-Patient Power Asymmetry

LANYUN ZHANG[1], JIANI ZHAN[1], VERENA KWOK WAI WAN[2], AND YANBIN WANG[1]
[1]Department of Industrial Design, College of Mechanical and Electrical Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing 210016, China
[2]Language Centre, Faculty of Arts, Hong Kong Baptist University, Hong Kong, China

Corresponding author: Lanyun Zhang (lanyunzhang@nuaa.edu.cn)

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ABSTRACT Online health consultation (OHC) platforms are widely used telemedicine services in China, and the number of active users has grown rapidly over the past 5 years. OHCs support various types of medical consultations, such as diagnosis and treatment advice, with registered physicians. However, the physician-patient power asymmetry that persists in face-to-face medical consultations has received little attention in telemedicine environments such as OHCs. This paper aims to explore how user interface (UI) design can approach the issue of power asymmetry in OHCs. First, a workshop (N=7), consisting of a focus group and a nominal brainstorming session, was organised to explore user needs, generate and discuss UI design solutions when considering power asymmetry. Study 2 is a Wizard-of-Oz design study (N=30), where the first step was to design a prototype of the OHC with two possible UI solutions (i.e. *informing* and *prompting*). It revealed that these solutions addressed power asymmetry and improved user perceptions of OHCs. This study contributes to the knowledge and implications of how UI design can address power asymmetry in OHCs.

INDEX TERMS Electronic healthcare, online health consultation platforms, user interfaces, physician-patient power asymmetry, user-centered design.

I. INTRODUCTION

In mainland China, the vast majority of patients go directly to hospitals for outpatient care regardless of the severity of their illness, except for emergencies. It results in the outpatient department constantly having a large number of patients. Additionally, the insufficient and unevenly distributed healthcare facilities have caused great pressure on top hospitals and adversely affected patient experience. It can take a patient two to three hours in a hospital before seeing an outpatient physician and the actual time spent on seeing the physician could be only 10 minutes. To address these problems, *online health consultation* (OHC) platforms have been introduced to allow patients to make enquiries about their health conditions and seek professional advice from physicians via the Internet [48]. At an individual level, OHC platforms provide patients with an efficient and effective way to access physicians worldwide for medical decision-making [75]; at a macro level, OHCs can potentially alleviate the increasing burden on healthcare systems with distributed medical resources [98]. The users of online health care systems have grown rapidly in China in the past 5 years, that they allow patients to see physicians in top hospitals without travelling and they also support both follow-ups after physical consultations and first-time consultations. The market of mobile healthcare services has been continuously expanding in China since 2015, with

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the number of active users seeking OHC services reaching over 15 million in June 2021 alone [97].

Despite these promising numbers and potentials, OHCs also have limitations. Discussions in the field of telemedicine express that little attention has been paid to the practice of patient-centred care in online services [8], [39], including addressing the issues of power asymmetry between physicians and patients. This power asymmetry has been found to be consistent in physician-patient interactions since the late 1960s [71]. Parson's observations of medical practice suggested that physicians exercised power over patients [70], and the quest for patient-centred medicine has continued to this day. Power asymmetry is currently being addressed through physician training that encourages more egalitarian styles of interaction in face-to-face encounters [24]. However, to the authors' knowledge, little attention has been paid to addressing the issue of physician-patient power asymmetry in telemedicine such as OHCs.

The main difference in managing power asymmetry between face-to-face encounters and online encounters is that in the latter, the user interfaces of telemedicine platforms act as a medium between patients and physicians, that they can be manipulated for specific purposes. User interface (UI) design has already been discussed in the science of improving user experience in telemedicine, such as transparency, understandability, and satisfaction [87], [101]. For example, UI design can help patients to understand the reasons why online symptom checkers made certain diagnoses, which improves transparency [87]. However, there is a lack of research on how UI design can help address the physician-patient power asymmetry in OHCs.

In this paper, we explored how UI design could address power asymmetry on OHC platforms. To answer this research question, we conducted two studies: First, we conducted a workshop consisting of a focus group and a nominal brainstorming session to explore user needs, generate and discuss UI design solutions when power asymmetry was considered. Second, based on the results of the first study, we conducted a design study with a Wizard-of-Oz user test to investigate how UI design affects power asymmetry and user perceptions in OHCs.

Our contributions to human-computer interaction are twofold: 1) we have provided insights into the role of UI design in addressing physician-patient power asymmetry in OHCs, which is very different from face-to-face health consultations, and 2) we have proposed design implications with empirical user test for addressing this power asymmetry in healthcare technologies, especially UI design in conversational interfaces.

Section II discusses the literature related to OHC platforms, physician-patient power asymmetry, UI design and its evaluation in telemedicine. Section III provides an overview of the design of this research, followed by Sections IV and V, which describe the two main studies in detail. Section VI discusses our findings in the context of UI design in telemedicine with future work. Section VII discusses the conclusion and limitations of this work.

II. RELATED WORK

A. ONLINE HEALTH CONSULTATION PLATFORMS

Online health consultation (OHC) is a type of mobile healthcare service that allows patients to consult with registered physicians about their individual medical problems via the Internet [100]. OHCs support various types of medical consultations (e.g. diagnosis and treatment advice) through conversational UIs [46]. First, among China's OHC platforms, the prevailing interaction mode of online consultations is dialogue-based, mainly text-based [11], [15]. Text- and dialogue-based consultations allow patients to express their thoughts without being interrupted by physicians or other patients [54], [95]. However, [96] found that physician-patient relationships could be hard to build with text-based consultations in OHCs. Therefore, dialogue-based consultations play an important role in OHCs, have positive influences on OHC experiences, and also show potentials with further research and design. Second, research on OHCs is limited, with previous studies mainly investigating the use of OHC platforms, such as interaction frequency and response speed [56], and the perceived factors that influence patients' desirability to use OHC platforms, such as ease of use [1], level of trust [88], and effectiveness [17]. A comprehensive research project was undertaken to examine the design processes and challenges associated with OHCs from the perspective of designers, with the aim of enhancing the digital patient experience [90]. However, most of the work has focused on the investigation of existing OHC services – how they are perceived by patients and how these perceptions affect patients' decision-making behaviour. There is a lack of research into how patients' interactions with OHCs can be empirically improved through design.

In the broader field of telemedicine, the literature has explored the design of telemedicine platforms to support shared decision-making (a process in which both the patient and physician contribute to the medical decision-making) to improve patient experience, but mainly for a specific health condition, such as depression [62], diabetes [35], infertility [44], or for a specific medical purpose, such as symptom reporting [78]. However, OHC platforms in mainland China are different that they support medical consultations across all departments [37], [45], [83]. To the authors' knowledge, there is little research exploring how the UIs of telemedicine platforms can be designed to mitigate a general issue of patient experience regardless of a specific health condition and possibly generalise to broader medical consultation scenarios on the Internet.

B. POWER ASYMMETRY BETWEEN PHYSICIANS AND PATIENTS

A critical issue that has been identified in patients' face-to-face interactions with physicians is the asymmetry of power

between physicians and patients [4], [6]. Power is pervasive and it can come from a variety of sources, making it difficult to describe accurately and clearly [32]. Power can be seen as the possession of valuable resources [30] or social resources, such as decision-making opportunities [49], or as the ability to influence others and change their attitudes and behaviour [82]. Research also suggests that all types of power share a common characteristic: dependence, and it is the degree of dependence between individuals that allows power to occur [28].

In this context of healthcare, patients are dependent on physicians for medical information, expert advice, test results, and interpretations of test results. A physician has the power to influence a patient's diagnosis, treatment, medication, and ultimately health status. Physician-patient power asymmetry can be caused by various factors, such as gender [22], the power to open human body and to prescribe potentially lethal substances [53], social economic status [12], and the access to sensitive personal information and examination results [53]. The concept and evidence of power asymmetry between physicians and patients have been extensively explored in the extant literature [4], [6]. Existing research has found a systematic pattern that show physician's exercise of power and authority over patients, from analysing their conversations. For example, physicians interrupt patients more than the reverse [93] and physicians decide which topics are relevant to the consultation [21]. On the other hand, there are a few studies that have explored various types of physician-patient relationships, such as patient-led titration [33] and consumer-directed health care [10]. Yet, physicians mainly play a paternalistic role during medical consultations in China, that patients are given little time and opportunities to express their thoughts [11]. Therefore, this research mainly focused on exploring this issue where physicians play the dominant role in OHCs, with possible solutions to address this power asymmetry.

Works in the fields of patient engagement [61] and shared decision-making (SDM) [38] have explored ways to address the power asymmetry between physicians and patients. For example, a model consisted of three components (i.e. *choice, option,* and *decision talk*) was proposed to guide the clinical practice of SDM [27], that can reduce the imbalance. However, the practices to reduce physician-patient power asymmetry in mainland China is very limited [43]. In this research, Study 2 adopted an assessment protocol of SDM practices to guide our design, further SDM-related information will be elaborated in section V-B2.

Moreover, most of the literature focuses on studying physician-patient power asymmetry in face-to-face encounters [6]. Little work has focused on the online environment or application design. Among the limited studies in virtual settings, [92] have found that online health information can mediate the asymmetric power balance between physicians and patients. More recent studies have shown that medical information found on the Internet only has the potential to narrow the gap between physicians and patients, but not to bridge it, and thus the power balance remains asymmetric [25]. Overall, these studies have mainly examined the manifestations of power asymmetry in both offline and online encounters, examined whether existing online information could mitigate physician-patient power asymmetry. However, they have not explored how patients' interaction with online platforms could be designed to empirically address this power asymmetry [99].

C. UI DESIGN OF TELEMEDICINE

When patients interact with OHCs, they do so through a user interface. UI design is considered important because existing research has (1) explored the design of telemedicine UIs for specific user groups, such as culturally specific groups [2] and elderly user groups [3]; (2) found that UIs significantly influence the user perceptions of a telemedicine APP, such as transparency [87]; and (3) explored different types of UI design in telemedicine, such as explainable UI and conversational UI [87] in the context of online self-checkers without interacting with human physicians. Our work is deeply rooted in conversational UI design [55], which combines human and artificial intelligence (AI). Conversational UIs refers to chatbots and different forms of assistants in AI assistants that are widely used in everyday life, such as assistants in cars and Amazon Alexa [77]. In this context of OHCs where patients interact with human physicians, little attention has been paid to addressing the issue of power asymmetry through UI design, despite the importance and enormous popularity of OHCs. This work aims to explore how conversational UI can be designed to approach the issue of physician-patient power asymmetry.

D. EVALUATION OF UI DESIGN IN TELEMEDICINE

To evaluate the UI design, a user-centred evaluation framework was adopted to measure physician-patient power asymmetry and user perceptions. User perceptions were investigated with six dimensions: satisfaction, effectiveness, transparency, learning, e-health literacy, and social presence.

Satisfaction and *effectiveness* are widely examined in digital tools, while *transparency* and *learning* are widely examined in mobile health systems. The choice of constructs and questions of these four dimensions was inspired by the work of [51], [52], [86], and [87].

First, *transparency* "requires full disclosure of all relevant information in a timely manner", and it serves as a vital tool for overcoming information asymmetries [29]. In the context of OHCs, transparency refers to an OHC's ability to help explain physician's decisions or recommendations.

Second, with regards to *learning*, adopted from [87], it refers to an OHC's ability to help the patients to gain health-related knowledge and health awareness about themselves.

Third, health literacy is defined by the World Health Organisation (WHO) in 1988 as "*the cognitive and social skills that determine the motivation and ability of individuals*

to gain access to, understand, and use information in ways that promote and maintain good health" [65]. *E-health literacy* is the equivalent to health literacy in the digital era. Existing works have found that individuals' education level, access to e-resources, level of health concern, level of training involved, the frequency of Internet usage [16], [69], [80], [89], [94] have significant impact on his/her e-health literacy. In the context of OHCs, e-health literacy reflects patients' abilities to use OHC platforms to seek and understand health information and make informed decisions to achieve better health outcomes. This study seeks to explore e-health literacy in OHCs and to examine the impact of UI design on patients' e-health literacy.

Fourth, *social presence*, defined as the "*degree of salience of the other person in the interaction and the consequent salience of the interpersonal relationship*" [81], has been widely discussed in the context of virtual environments, such as online shopping [59], online education [13], and website design [42]. Social presence is a central concept in interface design as researchers explore how media can be used to overcome physical distance and the lack of human touch [34], [40]. In the context of OHCs, social presence represents the warmth, sociability, and sense of human contact that patients experienced through the OHC platforms. This study aims to investigate whether and how the UI design of OHCs can affect patients' social presence.

III. RESEARCH DESIGN

To answer the research question of how UI design can address physician-patient power asymmetry in OHCs, two consecutive studies were designed. First, a workshop (including a focus group session and a brainstorming session) was held to explore user needs and generate UI design solutions for power asymmetry during OHC experience, and to allow discussion of the proposed solutions. This study was premised on the general recognition of exploring UI design to facilitate the experience of users' interaction with virtual systems. A workshop study is deemed appropriate to identify the role of UI design in users' interactions with OHCs and to generate possible solutions. Building upon the identified user needs and design solutions, Study 2 designed and tested how the UIs derived from Study 1 affected power asymmetry and user perceptions of OHCs. Knowledge about UI design for OHCs was extracted. Ethical approval was obtained for both studies before they were conducted and all participants have signed the consent forms with relevant information before the start of each study.

IV. STUDY 1: USER-ORIENTED OHC INTERFACE DESIGN

A. AIM

Study 1 aims to explore user needs and generate possible UI design solutions to address physician-patient power asymmetry in the use of OHCs, and how OHC interfaces can be designed to support end user values from a physician-patient power asymmetry perspective.

B. METHOD

1) STUDY DESIGN

A workshop, including a focus group session [72] and a nominal brainstorming session [41], was organised to identify evidence-based strategies associated with real OHC experiences, including but not limited to existing research and stakeholders' knowledge and past experiences. Focus group is a technique involving the use of in-depth group interviews in which participants are selected because they are a purposive sampling of a specific population [72], [85]. Nominal brainstorming, as the focus of this study, is a structured method for group brainstorming that encourages contributions from everyone and facilitates quick agreement on the relative importance of issues, problems, or solutions [74]. Relevant topics were identified, classified, and facilitated in this workshop by using pen, paper, post-it notes and a whiteboard.

2) PARTICIPANTS

A total of 7 participants took part in the workshop through word-of-mouth within the university and among friends. Six participants were regular users of OHC platforms (at least twice a year, consulting for themselves or their relatives), and at the same time they were professionally trained as either UI designers or user experience designers. All the OHC platforms they have used were in Chinese and registered in mainland China. One participant was a registered physician who had experience as a telemedicine professional on one of the Chinese OHC platforms. Details of participant information are shown in Table 1. The workshop was conducted on campus in a face-to-face format.

3) EXPERIMENT PROCEDURE, DATA COLLECTION

The workshop consisted of two sessions. In the first session (focus group), the aim was to share and identify the problem, question, or issue that was related to physician-patient power asymmetry in OHCs, and to stimulate discussion to understand user needs. So, participants were asked to recall their past experiences with any OHC platform as either a physician or a patient, paying particular attention to physician-patient communication; and select one of the experiences that could best reflect physician-patient power asymmetry. The participants were given 10 minutes and to use pen and paper to write down the symptoms, the diagnosis, and the agreed treatments and medications, if any. Each participant was given a further 10 minutes to verbally describe their experience, and then discussion was encouraged among the participants.

The second session (nominal brainstorming) aimed to generate solutions to the question: how can OHC interfaces be designed to address the issue of physician-patient power asymmetry? Participants were given 10 minutes to silently think of solutions or ideas that come to mind and write down as many as possible. Then, each participant shared one idea aloud in each round among the group members until all members had had their turns. The facilitator recorded each idea on

TABLE 1. Participant information of study 1.

No. of participants	Research domain is a mix of …	Organisation	Past OHC experience
1	User experience (UX) design, product design	Industrial design, NUAA	Yes (patient)
2	UX, human computer interaction, virtual reality	Industrial design, NUAA	Yes (patient)
3	Product design, ergonomics	Industrial design, UNNC	Yes (patient)
4	Visualisation, UI design	Art & design, NUAA	Yes (patient)
5	UI design, communication design, product design	Art & design, NUAA	Yes (patient)
6	UI and UX design	Design & creative art, LU	Yes (patient)
7	Thyroid cancer diagnosis, surgery, and treatment	Head and neck, CHJP	Yes (physician)

NUAA = Nanjing University of Aeronautics and Astronautics, UNNC = University of Nottingham Ningbo China, LU = Loughborough University UK, CHJP = Cancer Hospital of Jiangsu Province

a post-it and stuck them all on a whiteboard. A discussion of each idea was then arranged. Participants were encouraged to explain their ideas and to their opinions about the ideas of others.

4) DATA ANALYSIS

The data collected in the first session were not analysed and discussed in depth in this paper, as the participants' personal stories *per-se* and the manifestation of power asymmetry were not the focus of this study. The first session acted as a retrospective process to recall memories, explore user needs, and provide a basis for the next session.

The data collected in the second session were the ideas and solutions that could potentially address the power asymmetry. These qualitative data were sorted based on affinity diagramming [58], and further interpreted through thematic analysis [84]. The initial sorting was done *in-situ* at the end of the session with all participants presenting, as the researchers wanted to ensure that all the ideas and solutions were accurately recorded and interpreted.

C. RESULTS

The comments from the focus group session were classified into two themes: cognitive aspect and affective aspect of physician-patient power asymmetry in OHCs; while the ideas and solutions that emerged from the brainstorming session were grouped into two main themes: cognition-driven solutions and affect-driven solutions. Figure 1 illustrates the rationale and findings of Study 1. This section focuses on elaborating on the findings of the brainstorming session with brief explanations of the comments from the focus group session.

1) COGNITIVE ASPECT AND COGNITION-DRIVEN SOLUTIONS

The cognitive aspect of power asymmetry between patients and physicians is most often due to unequal levels of knowledge, experience, and expertise in relation to a diagnosis or treatment. One participant said that "*sometimes I don't understand what the physician means or why he/she makes a diagnosis. I think it's because I don't have the relevant information. What's obvious and straightforward to him (the physician) is not to me*". The essence of addressing this issue is to position the person seeking advice and guidance (e.g. the patient) as an expert in their own right: an expert in their body, views, concerns, goals, priorities, and preferences [26]. One possible approach to quickly turn a patient into an expert is to "*educate the patient whenever needed*" as one participant said. The cognition-driven solutions focus on providing patients with different types of relevant medical information that they may lack in a consultation. This is reflected in the cognitive mindset that sees accumulated knowledge and past experience as key elements [47]. In the context of OHCs, one participant explained that "*whenever I came across any medical terminology or something the physicians said that I was not sure about, I would look it up. I would go online and find out for example, what it means and what to do with it*". It was then suggested that "*information about a disease and related medical knowledge can be helpful during the consultation…links can be included in OHC interfaces to direct patients to this information*". Based on this empirical finding, one possible solution – *informing* was derived to explore how UI design can support cognitive needs. Details of *informing* and further user test are elaborated in Section V.

2) AFFECTIVE ASPECT AND AFFECT-DRIVEN SOLUTIONS

Although the cognitive aspect tends to be knowledge-driven, this does not mean that power asymmetry and possible solutions are exclusively based on rational thinking. The affective aspect of power asymmetry is reflected in the fact that patients are anxious, uncertain, vulnerable, and in desperate need of a dependent, who is likely to be the physicians in most cases. Although medical decision-making should be evidence-based, it is equally important to acknowledge that every person's decision draws deeply on narratives, personal connections, intuitions, emotions, and relationships [26]. For example, one participant expressed that "*for me, I would like the physicians to initiate any questions about my symptoms, because I always feel anxious when I go to the physician and worry that I might forget something important*". A patient, who is in a vulnerable state and with less power, calls for greater attention to the emotional and relational dimensions of care [26]. Drawing on the concept of affect in psychology [5], feelings and emotional bonds exist between patients and physicians should also be facilitated. The affect-driven solutions [23] focus on encouraging physicians to involve

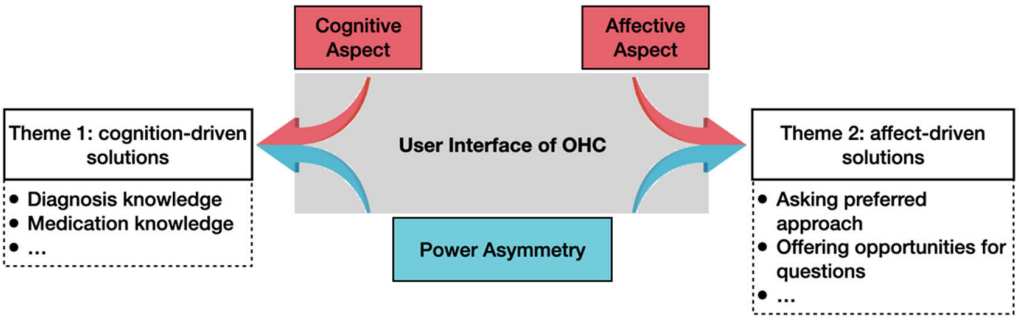


FIGURE 1. Illustration of the rationale and findings of Study 1.

patients in the medical decision-making process, where both parties should actively discuss, listen, and work together to reach a decision. The essence of affect-driven solutions is that physicians are the initiators of communication, proactively creating an environment that is caring and willing to listen. "*It would actually help a lot if the physician asked first – if I want to take oral or topical medication, rather than just giving it to me and seeing if I accept it*", said one participant. Based on this empirical finding, one possible solution – *prompting* was derived to explore how UI design can support affective needs. Details of *prompting* and further user test are elaborated in Section V.

V. STUDY 2: WIZARD-OF-OZ USER TEST

A. AIM

Study 2 aims to explore how cognition-driven and affect-driven solutions affect the user perceptions in OHCs. A Wizard-of-Oz experiment was conducted in this study. A prototype of OHC user interfaces was developed, based on the findings from Study 1, to allow the Wizard-of-Oz experiment.

B. OHC INTERFACE DESIGN

We decided to replicate the interfaces of a popular Chinese OHC platform and embed our designs into the interfaces for user testing. Firstly, since all OHC platforms in China share an overall resemblance with minor differences, Yilu (医鹿), as a leading OHC mobile application in China, was chosen. The mock-up APP was developed using Adobe XD with a fixed symptom (acne) as the topic of the consultation. Acne was chosen in this study because skin conditions were frequently consulted on OHC platforms during Covid-19 in China [91]. Secondly, a closed Wizard-of-Oz paradigm [73] was chosen because none of the "wizards" in this study had a medical background to allow for an open paradigm. Thus, the content of the OHC conversations was predetermined and fixed for all participants. In our replica, we added two new designs, *informing* and *prompting*, into the conversational interfaces between physicians and patients (see examples in Figure 2 and 3). *Informing* and *prompting* were designed based on the empirical findings of cognition- and affect-driven solutions respectively in Study 1.

1) DESIGN 1: INFORMING

This type of design offers medical information and knowledge in the form of an encyclopaedia via a link during the conversation (see an example in Figure 2). Medical terminologies can be confusing to patients, who might misunderstand their meanings or have never heard of them. *Informing* provides a link, automatically generated by the system through keyword detection of a conversation between physician and patients. Patients can choose to click the link for information or ignore it. The link provides patients with relevant information about the medical terms mentioned by the physicians in the OHCs, such as causes of a disease, side effects of a medicine, etc.

2) DESIGN 2: PROMPTING

This type of design gives patients the opportunity to express their thoughts, including five prompted parameters: *understanding, concerns, expectations, preferred treatment, and questions*. An example is shown in Figure 3. The five parameters are taken from the OPTION12 (12-item Observing Patient Involvement scale) items, see highlighted in Appendix A. OPTIOIN12 is an assessment protocol of SDM practices, that measures the patient involvement in SDM by rating the physician's level of expertise for 12 key 'patient-involving' behaviours [19]. Shared decision-making (SDM) is a key approach to patient-centred care, that it facilitates the physician-patient communication to co-produce an evidence-based, sensible, and feasible response to the problematic situation of the patient [38]. As participants in Study 1 suggested different topics for the prompt, OPTION12 was adopted as a reference to structure these topics. The five items of OPTION12 were selected because they focus on measuring whether a physician is proactively engaging in the interactions with a patient. In the OHC interface design, links are displayed to prompt patients to indicate whether they *understand*, whether they have *concerns*, what their *expectations* are, what types of *treatment* are accepted, and what *questions* they have. Then, the physician in consultation responds according to the patients' feedback on the prompts. The links provide open-ended or multiple-choice questions, and patients can choose to click on the links or ignore them.



FIGURE 2. An example of *informing* – providing relevant information about the diagnosis.



FIGURE 3. An example of *prompting* – asking patients' preferred approach.

C. METHOD

1) STUDY DESIGN

Independent variable was represented by four designs of OHC platforms: baseline (no added design), *informing*, *prompting*, and *informing & prompting*. These designs were generated based on the results of the workshop elaborated in Study 1.

Dependent variables. This study examined how OHC designs influenced power asymmetry and user perceptions. Firstly, *power asymmetry* is hard to measure because power itself (as a concept in social sciences) is hard to measure as it takes different forms [20]. The power asymmetry between a physician and a patient during a health consultation is often assessed by analysing real-time consultation recordings of the consultation, such as conversation analysis [50], but rarely by self-reported methods such as surveys and interviews [71]. In this study, questions about power asymmetry were not asked directly in the survey or the interviews. Rather, emergent thematic analysis [66] was used to derive participants' views and opinions on power asymmetry. Secondly, user perceptions were measured with six dimensions. (1) Adapting a study on online symptom checkers [87], this study measured the *effectiveness*, *transparency*, *learning effect*, and *satisfaction* of OHC designs. (2) Very few studies have explored *e-health literacy* in OHCs, and they have mainly focused on creating and testing conceptual models [102]. This study seeks to examine how the UI of OHCs influences users' e-health literacy by using a revised eHEALS (the eHealth Literacy Scale) – access, appraise, and use [64], [65], [67]. (3) This study aims to investigate whether UI design can influence patients' social presence by adopting a revised scale in Lu et al.'s work [57] – social presence of web and social presence of interaction [34], [57].

2) PARTICIPANTS

This type A total of 30 participants were recruited through social media and word-of-mouth in China. To be qualified for this study, participants had to be between 18 and 35 years old (people within this age group are the most frequent users of OHCs in China [91]). Each had a laptop computer with Internet capability so that the researchers could communicate with them if they were unable to meet the researchers in-person due to COVID-19. Eight participants took part in the study via an online conferencing platform – Tencent Meeting. Participant information is summarised: (1) 15 females and 15 males (2) 22 postgraduate students and 8 non-students (3) 18 with past OHC experiences, 4 with past online medicine ordering experiences, 2 with past online medical information browsing experiences, and 6 with none of the above. Participants were compensated for their time to motivate the participation.

3) EXPERIMENT MATERIAL AND PROCEDURE

At this stage, the research was not ready for a full investigation with a fully functional App and real patients. Therefore, this study took a Wizard-of-Oz approach to user testing. A prototype of OHC interfaces with new features was created using Adobe XD, with a fixed symptom (acne) as the topic of the consultation. Acne was chosen in this study because skin conditions were frequently consulted on OHC platforms during Covid-19 in China [91]. The content of the OHC conversations was predetermined and fixed for all participants. To 'recreate' and simulate an authentic experience with OHC, this study began by describing the predetermined scenario to each participant: ''you have noticed that you have had spots on your cheeks for a while, and you decide to consult a physician using an OHC platform''. Then, the study introduced the OHC prototype and required the participants to click the ''send'' button to submit each pre-written message. Four versions of the OHC experience were created: 1) baseline experience, a standard OHC platform, 2) *informing* experience, 3) *prompting* experience, and 4) *informing & prompting* experience. We followed a within-subjects design, asking all participants to use the baseline version of the OHC prototype and two random versions of the other three (see Figure 4. a). Participants were asked to fill out a survey after completing each version (see questions in Appendix B). At the end of the study, 15 of the participants were invited to participate in a semi-structured interview, structured as follows: overall opinion, user perceptions, e-health literacy, and social presence.

4) DATA COLLECTION AND ANALYSIS

A mixed-method approach was used to address the research questions, combining survey measures and semi-structured interviews. The quantitative survey collected participants' subjective perceptions of their individual experiences with the given OHC designs. The dependent variables, measured via the survey, were represented by a 7-point Likert scale (Strongly disagree to Strongly agree). IBM SPSS software was used to analyse the quantitative data (see datasets used for analysis in Figure 4. b, c, and d). Wilcoxon sign tests were used to analyse the data since the sample sizes were too small for paired-sample t-tests. A follow-up interview was conducted to elicit the opinions of the participants to gain an in-depth understanding. As the interview transcripts were qualitative in nature, a thematic analysis [9] was used, guided by a combination of a *priori* coding and emergent coding [76]. The *a priori* codes consisted of: (1) power asymmetry, (2) effectiveness; transparency; learning; satisfaction; e-health literacy – access, appraise, use; and social presence – web, interaction.

D. RESULTS

1) POWER ASYMMETRY

Analysis of the interview data revealed that all participants noticed the designs – *informing* and *prompting* and had positive views towards these two designs. For example, regarding *informing*, participants said that: ''*it helped me understand her (physician's) diagonosis*'', ''*I could actually check my skin condition with this information, see if it matched the physician's words, and maybe self-check later*'', ''*it alleviated the odd feeling of compeletly being led by the phycisian during the consultation process*''. Regarding *prompting*, participants said that: ''*I liked it because I could see my options, eliminated those that I might struggle, and the physician would consider my preference*'', ''*I felt like I finally had some power in my own treatment plan, rather than passively accepting the physician's*''.

Then, the analysis mainly focused on the issues raised in relation to physician-patient power asymmetry. Firstly, the most frequently mentioned issue was ''why I was instructed to take THE medicine''. The participants showed a strong dependency on physicians to provide them with more information and details about why certain medicines were prescribed. Of the 6 participants who mentioned this issue, 4 stated that the designs in *informing* and *prompting* could effectively address this issue. Secondly, the issue of ''why I was diagnosed with acne'' was raised by 4 participants, which exemplified the rationale why patients are dependent on physicians to explain the cause, nature, and manifestation of a condition or disease. 3 out of 4 participants expressed that the issue was not properly addressed by the designs, as one participant questioned, ''*I don't understand why the physician diagnosed me with acne. She just made the diagnosis without asking me any questions*''. This implies that the physician had the ultimate control over the explanation of a diagnosis and that the designs of *informing* and *prompting* played a marginal role. Thirdly, ''whether I learned anything from this OHC session to help with future self-diagnosis'' was mentioned regarding the design of *informing*, indicating that *informing* empowered patients with relevant medical knowledge so that patients could, to some extent, track and monitor their own health. Fourthly, one participant raised the issue of ''personalisation of OHC services'', that

	Baseline	Informing	Prompting	Informing & Prompting
Group 1 (n=10)	10	10	10	0
Group 2 (n=10)	10	10	0	10
Group 3 (n=10)	10	0	10	10

a. Distribution of participant groups

	Baseline	Informing	Prompting	Informing & Prompting
Group 1 (n=10)	10	10	10	0
Group 2 (n=10)	10	10	0	10
Group 3 (n=10)	10	0	10	10

b. Datasets used to analyse *Informing*

	Baseline	Informing	Prompting	Informing & Prompting
Group 1 (n=10)	10	10	10	0
Group 2 (n=10)	10	10	0	10
Group 3 (n=10)	10	0	10	10

c. Datasets used to analyse *Prompting*

	Baseline	Informing	Prompting	Informing & Prompting
Group 1 (n=10)	10	10	10	0
Group 2 (n=10)	10	10	0	10
Group 3 (n=10)	10	0	10	10

d. Datasets used to analyse *Informing & Prompting*

FIGURE 4. Participant groups and datasets for analysis.

"*I liked the pop-up cards to personalise my treatment plan. It is a form of freedom from her (the physician's) power*", indicating that tailored information and personalised features can achieve patient empowerment. Fifthly, "whether OHC supports patient autonomy" was discussed by one participant. He/she praised that "*the new designs (informing and prompting) kind of allowed me to control my life as I wish, as I have the autonomy*".

2) SURVEY ANALYSIS OF USER PERCEPTIONS

The quantitative survey results showed that the UIs with *informing* and *prompting* significantly increased user perceptions in multiple aspects. In Figure 5, 6, and 7, Likert scale measured from 1 "strongly disagree" to 7 "strongly agree"; triangles in represent the median responses of each measured variable; circles represent individual responses. All measurements of user perceptions are significantly above the median (Mdn = 4, $p < 0.05$) for the UIs of *Informing*, *Prompting*, and *Informing & Prompting*, with the exception of two measurements for *Informing* (i.e. social presence of Web and social presence of interaction).

(1) *Effectiveness* measured whether the OHC platform supported the users in making appropriate/well-informed decisions. Results revealed that only the OHC with *informing & prompting* ($Z = 2.516$, $p = 0.012$) scored significantly higher than the baseline version in terms of effectiveness. (2) *Transparency* assessed whether the OHC platform explained why diagnoses or decisions were made. Results demonstrated that the designs with *informing* ($Z = 2.442$, $p = 0.015$), *prompting* ($Z = 3.248$, $p = 0.001$), and *informing & prompting* ($Z = 2.977$, $p = 0.003$) were seen as significantly more transparent than the baseline version. (3) *Learning* measured the learning effects after using OHC platforms. Results showed that the OHCs with *prompting* ($Z = 3.357$, $p = 0.001$) and *informing & prompting* ($Z = 2.920$, $p = 0.004$) were perceived as significantly better than the baseline version. (4) *Satisfaction* evaluated the overall enjoyment and willingness to reuse OHCs in the future. Results indicated that the OHC designs with *informing* ($Z = 2.530$, $p = 0.011$), *prompting* ($Z = 2.508$, $p = 0.012$), and *informing & prompting* ($Z = 3.225$, $p = 0.001$) were viewed more significantly satisfied than the baseline version.

(5) *E-health literacy*. 1) *Access* evaluated whether the users could get access to health-related information via OHC platforms. Results showed that the OHCs with *prompting* ($Z = 2.165$, $p = 0.030$) and *informing & prompting* ($Z = 3.296$, $p = 0.001$) scored significantly higher than the baseline version which means it was perceived as more accessible. 2) *Appraise* measured whether the users could understand and evaluate the content provided by OHC platforms. Results demonstrated that only the OHC with *informing & prompting*($Z = 3.038$, $p = 0.002$) were seen as significantly better than the baseline version. 3) *Use* accessed whether the users felt confident to draw on the OHC information to solve current medical problems. Results suggested that only the OHC with *informing & prompting*($Z = 3.715$, $p = 0.001$) was scored significantly higher than the baseline version.

(6) *Social presence*. 1) *Social presence of Web* evaluated whether the users felt a sense of human contact when using OHC platforms. Results revealed that the OHCs with *informing*($Z = 2.489$, $p = 0.013$), *prompting*($Z = 3.224$, $p = 0.001$), and *informing & prompting* ($Z = 3.787$, $p = 0.001$) were regarded as better than the baseline version. 2) *Social presence of interaction* evaluated whether the users felt as if they were interacting with a physician in person when using OHC platforms. Results showed that the OHCs with *prompting* ($Z = 3.145$, $p = 0.002$) and *informing & prompting* ($Z = 3.162$, $p = 0.002$) were viewed significantly better than the baseline version.

3) NEGATIVE COMMENTS OF USER EXPERIENCE

The second analysis of the interview data focused mainly on the negative comments of user perceptions, e-health literacy, and social presence, to guide the design of OHC platforms. Resonating with the survey results, comments collected in the follow-up interviews were mostly positive. See Table 2, a total of 91 negative comments were collected and 20 sub-codes were derived. All these sub-codes were labelled with cognitive and/or affective dimensions. The criteria of

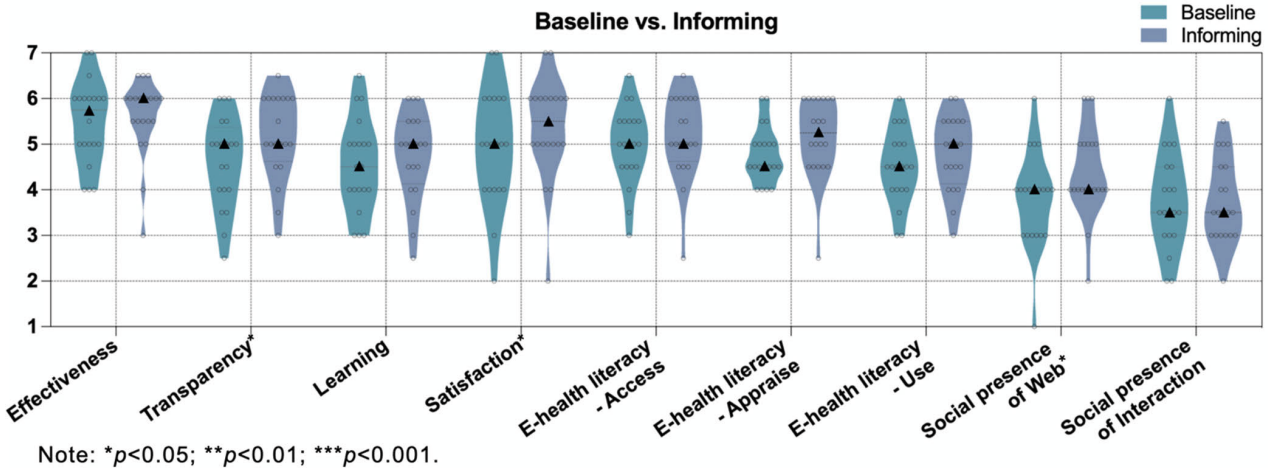


FIGURE 5. User perceptions between baseline OHC and OHC with *informing*. (* p<0.05, ** p<0.01, *** p<0.001).

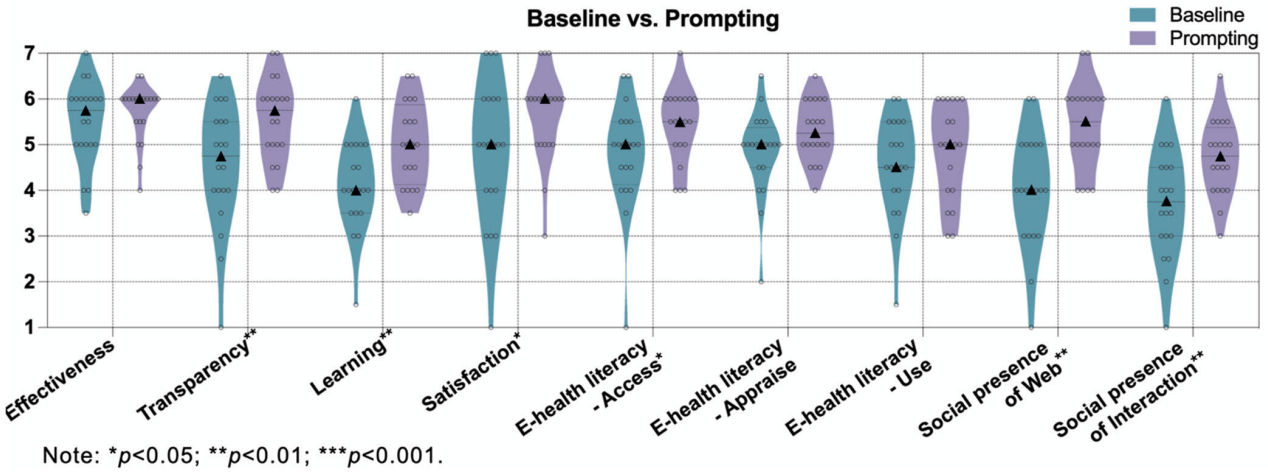


FIGURE 6. User perceptions between baseline OHC and OHC with *prompting*. (* p<0.05, ** p<0.01, *** p<0.001).

categorising criticality are: ''high'' when number of cases coded is greater than 5, ''median'' when number of cases coded is equal to or greater than 2 and equal to or smaller than 5, ''low'' when number of cases coded is smaller than 2.

In terms of *Satisfaction*, 16 negative comments were recorded and coded into three categories – lack of personalised information, trust, and privacy issues. The first two categories were mentioned most frequently by the participants. Regarding the lack of personalised information, one participant expressed that ''*I did not feel like being given enough information to target my problem*'', ''*the information provided can be easily searched myself*''. As for trust issues, people voiced concerns whether the physician's advice could be taken with full confidence, as one participant said ''*(the physician's) conclusions based solely on a few photos and questions were less convincing*''.

A total of 9 negative comments regarding *Effectiveness* were collected, which were put into three categories – inadequate medical information provided by the physician, misunderstanding between patients and physicians due to the limited communication online, and incomplete patient information asked by the physician.

The lack of *Transparency* perceived by some participants was mainly due to two reasons. First, diagnoses made seem to be swift. For example, one comment read: ''*the diagnosis and prescribing process seems to be quite fast that the physician should have asked me more questions*.'' Second, information provided by the physician was inadequate to explain some fundamental questions, such as the cause of the disease. One participant noted that ''*the physician did not explain what exactly caused my acne''* and that the physician's explanation *''was very superficial and only just scratched the surface*''.

In terms of *Learning*, two participants pointed out that there was not enough information to support the depth and breadth of learning. Meanwhile, one participant, who lacked the desire to learn, felt that the encyclopaedia-like medical information was unnecessary and redundant, ''*In my opinion, physicians don't need to educate patients as students. As a*

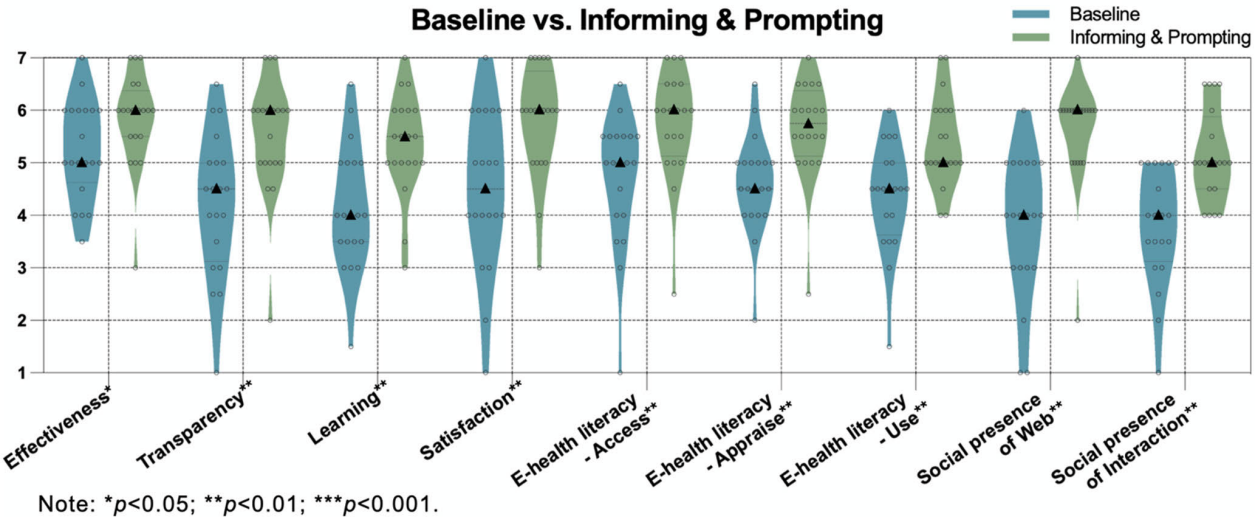


FIGURE 7. User perceptions between baseline OHC and OHC with *informing & prompting*. (* p<0.05, ** p<0.01, *** p<0.001).

patient, I will be happy as long as my symptoms are taken care of."

To improve e-health literacy, the ability to get access to online health information is essential. The analysis showed that *Access* received 3 types of negative comments: cognitive overload due to reading lengthy text messages, too many interactive features, and the lack of personalised interactive features. The latter two shared a common view – they were happy to take part in the medical decision-making process but less happy with the approaches. One participant suggested that "*I don't think the interactive features are necessary, such as the pop-up card to fill out my medical information. It would be better if the physician asked me the questions directly*", while another mentioned that "*I liked the pop-up cards, but I want to be able to skip them if I'm not interested*". Regarding *Appraise,* the participants attributed the main cause of the difficulty faced when assessing the consultation outcomes to the limited and remote communication with the physicians. As for *Use,* two types of negative comments were prominent: (1) difficult to use consultation outcomes due to the insufficient depth of information provided, and (2) not confident enough to take the physician's advice.

There were 15 negative comments related to Social Presence, in which the majority of participants indicated, "*a feeling of communicating with a robot*" rather than a real person. Some participants stated that the physician "*seemed to follow a script when asking questions, like a chatbot*". In addition, 5 participants felt that they were unable to picture the image or sense the attitude of the physician. Also, two contradictory views were recorded. On the one hand, the physician was viewed as being too formal to have a sense of human touch that "*everything about her made her surreal*". On the other hand, the physician was viewed as being too casual and friendly to be a professional that "*I could hardly feel her professionalism*".

VI. DISCUSSION

We explored how UI design could address physician-patient power asymmetry in OHCs with a two-phase approach: (a) a workshop to explore user needs and generate possible UI design ideas; (b) a design study to explore how UI design can influence power asymmetry and user perception during patients' interaction with OHCs. This section discusses the findings of this work in terms of theoretical contributions, practical implications, user experience of online patients, and possibility of generalising to broader user groups.

A. A COGNITIVE AND AFFECTIVE APPROACH TO ADDRESSING POWER ASYMMETRY IN OHCs

The issue of physician-patient power asymmetry was looked into from a cognitive and affective perspective, and the designs of *informing and prompting* were derived accordingly. The results suggested that the issue of power asymmetry was highlighted by the designs. Firstly, [25] found that health information through the Web mainly influenced the communication between patients and physicians, that the power balance remained asymmetric. The results of this study mainly reflected the above finding, so it is worth exploring whether cognitive approach (such as informing) can address the information asymmetry between physicians and patients, to eventually resolve the power asymmetry. Secondly, patients are often encouraged to be active rather than passive in their own health care, so it is equally important for physicians to connect with patients on a personal level to foster patient engagement [31]. In medical practices, a physician's verbal communication choices significantly impact patient participation, which is related to power asymmetry [2002]. In this study, *prompting* was designed to "nudge" patients to express their own opinions via text messages, which was found to have positive feedback regarding power asymmetry. However, some participants found the

TABLE 2. Results of the interview data analysis (negative comments only).

User perceptions		[category] Sub-codes	Number of Cases Coded	Criticality Level
Satisfaction		[cognitive] Lack of personalised information.	7	High
		[cognitive, affective] Trust issues.	7	High
		[cognitive, affective] Privacy issues.	2	Low
Effectiveness		[cognitive] Inadequate medical information provided by the physician.	4	Medium
		[cognitive] Misunderstanding between patients and physicians.	3	Low
		[cognitive] Incomplete patient information asked by the physician.	2	Low
Transparency		[cognitive, affective] Diagnosis seems to be swift.	6	High
		[cognitive] Inadequate professional information provided by the physician to explain the cause of the disease.	5	Medium
Learning		[cognitive] Not enough information to support depth and breadth learning.	2	Low
		[cognitive, affective] Encyclopedia-typed medical information is too much (due to the lack of desire to learn).	1	Low
E-health Literacy	Access	[cognitive] Cognitive overload due to reading lengthy text messages.	2	Low
		[cognitive, affective] Too many interactive features that make things complicated.	2	Low
		[cognitive, affective] Lack of personalised interactive features.	2	Low
	Appraise	[cognitive] Difficult to assess the consultation outcomes.	6	Medium
	Use	[cognitive] Difficult to use consultation outcomes due to the insufficient depth of information provided.	3	Low
		[affective] Not confident enough to take the physician's advice.	1	Low
Social Presence	Social Presence of Interaction	[affective] Feel like communicating with a robot.	8	High
		[affective] Unable to picture the image or sense the attitude of the physician.	5	Medium
		[affective] Too formal to have a sense of human touch.	1	Low
		[affective] Too casual and friendly to be professional.	1	Low

prompting questions to be unnatural. So, it still remains unknown whether there are other ways to "nudge" without being robotic?

B. INFORMING OR PROMPTING OR BOTH?

Three OHC designs were tested quantitatively in this study through surveys. Informing, inspired by cognitive aspect, provided links to explain jargons used during consultations, specifically in terms of medical diagnoses and medications. Our results showed that *informing* enhanced user perceptions but only to a limited extent. *prompting*, inspired by affective aspect, showed significant impact on user perceptions but no significant impact on e-health literacy. Our findings indicated that affective and emotional interventions during OHCs could not enhance the level of e-health literacy significantly in a short period of time. We believed that e-health literacy could only be improved through cognitive interventions such as filling knowledge gaps via attending educational classes. Lastly, *informing& prompting* were shown to have significant influences on all aspects of user perceptions. In certain aspects, the design supported both cognitive and affective aspects (*informing& prompting*) outperformed the sum of the designs supported each individually (Informing, *prompting*). For example, Effectiveness and Use showed no improvement in either *informing* or *prompting* but showed significant improvement in *informing& prompting*.

C. TRICKY E-HEALTH LITERACY?

E-health literacy is "the capacity to obtain, process, understand, and use" online information in a health-related context, with the cognitive ability and social skills to use technology [79], [80], [89]. Some literature described one's health literacy as static (relatively fixed and consistent) that it would be likely to remain almost unchanged during his/her adulthood if there were no intensive interventions made, such as long-term education [7], [36]. However, some literature argued that it was dynamic as "people develop to use health information" [102]. Although one's health literacy can vary as experience with health circumstances and choices changes, the literature has not made clear about what may change and the extent of such change. Our findings indicated that perceived e-health literacy [64] could be improved over a short period of time via changing OHC interface features. However, it remains unknown whether performed e-health literacy [14], [64] can be improved via the same design simultaneously.

D. HOW TO DESIGN AND MANAGE INTERACTIVE FEATURES?

The essence of patient-centred care is to engage patients in the medical decision-making process, which used to be dominated by physicians alone in paternalism. Both *informing and prompting* have a few interactive features, such as using pop-up cards to obtain personal medical information and preferences. Interestingly, results of this study revealed

TABLE 3. Observation items in OPTION12.

Item	Referred to as	Behaviour Description
1	Identifying problem	The physician draws attention to an identified problem.
2	Explaining equipoise	The physician states that there is more than one way to deal with the identified problem.
3	Assessing preferred approach	The physician assesses patient's preferred approach to receiving information to assist decision making (e.g. discussion in consultations, read printed material, assess graphical data, use videotapes or other media).
4	Listing options	The physician lists 'options', which can include the choice of 'no action'.
5	Explaining pros and cons	The physician explains the pros and cons of options to the patient.
6	Exploring expectations	The physician explores the patient's expectations (or ideas) about how the problem(s) are to be managed.
7	Exploring concerns	The physician explores the patient's concerns (fears) about how the problem(s) are to be managed.
8	Checking understanding	The physician checks that the patient has understood the information.
9	Offering opportunities for questions	The physician offers the patient explicit opportunities to ask questions during the decision-making process.
10	Eliciting preferred involvement	The physician elicits the patient's preferred level of involvement in decision making.
11	Indicating need for decision	The physician indicates the need for a decision-making stage.
12	Short title of article	The physician indicates the need to review the decision.

Highlighted: observation items that physicians encourage patients to express their opinions.

TABLE 4. Survey questions.

DVs	Factors	Questions
	Effectiveness	Q1: I find that the App helped me make medical decisions.
		Q2: I find that the App is useful in making medical decisions.
	Transparency	Q3: I understood why medical decisions were made during the consultation.
		Q4: I obtained health information that explained the medical decisions.
	Learning	Q5: I learned knowledge about my conditions and the diagnosis.
		Q6: I learned knowledge about the treatment to the diagnosis I had.
	Satisfaction	Q7: I am satisfied with the App.
		Q8: I will use the App again if needed.
		Q9: I would like to recommend this App to my friends if needed.
e-Health Literacy	Access	Q10: I think that I can get access to health information via this App.
		Q11: I think that I know how to get useful health information via this App.
	Appraise	Q12: I think that I can understand the health information in this App.
		Q13: I think that I can evaluate the usefulness of the information in this App.
	Use	Q14: I think that I know how to use the information in this App to help myself.
		Q15: I think I feel confident in using the information in this App to make health decisions (i.e. self-diagnosis).
Social Presence	Social presence of Web	Q16: There was a sense of human contact when using the App.
		Q17: There was a sense of human warmth when using the App.
		Q18: There was a sense of sociability when using the App.
	Social presence of interaction	Q19: I think I could make sense of the attitude of the physician by interacting with him/her via the App.
		Q20: I think I could imagine how the physician may look like by interacting with him/her via the App.

that all participants were happy to take part in these activities, but a few were less satisfied and offered some alternative suggestions towards the design of these interactive features. We believe that the lack of interactive features might reduce patient-physician engagement, while too many or poorly designed interactive features would increase users' cognitive

load that weaken the usability of OHCs. This may be caused by the varied abilities of individuals using these features and their personal needs.

E. SOCIAL PRESENCE ON OHC PLATFORMS

Social presence is one of the essential elements to assess the design of an online platform pertaining to user's perception of reality within a computer-generated context [42], [57], [63]. Nowadays, the language and emojis used for customer services are widely studied to further facilitates social presence [57]. For example, chatbots automatically use phrases like "*Hi, Jane. What can I help you with?*" to show friendliness when welcoming customers. In this study, when real physicians used similar language to show their care, the participants immediately associated this kind of language with chatbots, which had a considerably negative influence on their social presence.

F. GENERALISING TO OTHER USER GROUPS?

Although this study concentrated on the prevalent OHCs in China, it is worthwhile to consider the potential for extrapolation to other user groups. (1) The ability to generalise to other domains geographically. Globally, the telemedicine market is expected to grow 23.2% annually until 2028, with an estimated market value of USD 285.7 billion by 2028 [68]. The telemedicine landscape is witnessing a surge in mobile applications. An analysis produced by Orangesoft indicates that conversational UI such as secure messaging is supported by most applications [68]. It can therefore be reasonably inferred that the results of this study can be generalised to other domains globally. (2) The capacity to satisfy the requisite specifications for particular medical conditions. OHCs for specific medical conditions are commonly used, mostly focusing on non-emergency conditions, such as for mental health, birth control, acne, and migraines. Furthermore, conversational UI can be observed in such applications, and the results of this study may potentially enhance the patient experience. (3) The ability to generalise to other stakeholders in OHCs. This research mainly focused on patients and the potential for UI design to address the power asymmetry between physicians and their patients. It would be equally beneficial to examine the experience of physicians in OHCs and investigate how UI can facilitate power symmetry in their work.

VII. LIMITATION AND FUTURE WORK

This study contributes to the knowledge of the role of conversational UI design in physician-patient power asymmetry in telemedicine domain with empirical evidence. However, there are some limitations. First, the cognitive and affective approaches need further exploration regarding their dimensions and depths, and we have not fully explored them in different empirical settings. For example, cognitive aspects might further include attention and memory, and the implications for design have not been fully explored in this work. Second, this study did not investigate whether different combinations of cognitive and affective designs could influence user experience differently. A third limitation of the study is the relatively small sample size, which may have an impact on the results. It would be beneficial for future research to expand the sample size and include more in-depth questions. Fourth, our work only looked at perceived e-health literacy and examined how it was influenced by the OHC designs. It lacks the research on whether the results will remain unchanged when performed e-health literacy is measured or a longitudinal study is carried out. Fifth, future work could explore how practitioners could better the design of interactive features on OHC platforms and how patients can fully utilise these features. Sixth, future work could further explore the language used on OHC platforms to practise patient-centred care, such as the use of tones and anthropomorphism. Seventh, the contribution of this work is derived based on the limited research methods to address and evaluate power asymmetry, further empirical investigations are encouraged. Eighth, a further avenue for future research would be to investigate the potential for generalisation to different user groups. This could include patients in different domains and geographical locations, as well as patients with varying medical conditions. Ninth, this work focused mainly on text and picture messages, whereas there are other forms of information available on OHC platforms.

VIII. CONCLUSION

This work explored how UI design could approach physician-patient power asymmetry in OHCs. Specifically, we first conducted a workshop to explore the user needs and to generate design solutions. We found two categories of user needs: cognitive and affective aspects, and generated a set of cognition-driven and affect-driven solutions. Then, a design study was conducted to firstly design and develop a prototype incorporating the cognition- and affect-driven UI solutions, and secondly to test how the solutions influenced patients' interactions with OHCs. Our results showed that the power asymmetry between physician and patient could be addressed through the design of *informing and prompting*; *informing* improved user perceptions but only to a limited extent; *prompting* had a great impact on user perceptions but had a limited impact on e-health literacy; *informing*& *prompting* had significant impacts on all aspects of user perceptions. The interview data were largely consistent with the survey results. Negative interview comments were further analysed to reflect on how OHC interfaces might be designed.

APPENDIX A
OBSERVATION ITEMS IN OPTION12 (Couët et al., 2013)

See Table 3.

APENDIX B
SURVEY QUESTIONS

To what extent do you agree with the following statements (1- strongly disagree to 7- strongly agree): See Table 4.

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LANYUN ZHANG received the B.S. degree in electronics engineering from Nanjing University of Science and Technology, China, in 2010, the M.S. degree in signal processing from The University of Manchester, U.K., in 2011, and the Ph.D. degree in human–computer interaction from the University of Nottingham, U.K., in 2018. From 2018 to 2020, she was a Research Associate with the Design School, Loughborough University, U.K. Since 2020, she has been a Lecturer with the Department of Industrial Design, Nanjing University of Aeronautics and Astronautics. Her research interests include human–centered design, human–computer interaction, exploring how technology can play a role in creating a better user experience, and how technology should be designed to trigger behavior change and shape society. She was a recipient of the Shuangchuang Award of Jiangsu Province for Excellence, China, in 2021.



JIANI ZHAN received the B.S. and M.S. degrees in industrial design from Nanjing University of Aeronautics and Astronautics, China, in 2021 and 2024, respectively. Her research interests include human–computer design, human–centred design, how design interventions can enhance user experiences across various domains, and how design can help bridge the gap between users and technology.



VERENA KWOK WAI WAN received the master's degree in applied linguistics from University of Nottingham (Ningbo). She is currently pursuing the Ph.D. degree in education with the University of Bristol, U.K. She studied and worked at two universities in Mainland China before moving back to Hong Kong. Since 2020, she has been a Teaching Assistant with the Language Centre, Hong Kong Baptist University. Her research interests include education, language and culture, human behavior, applied linguistics, and intercultural communication. She received a Team Teaching Award while working on the English Across the Curriculum (EAC) project, from 2022 to 2023.



YANBIN WANG received the B.S. degree in aircraft manufacturing engineering from Beihang University (BUAA), Beijing, China, in 2004, and the M.S. and Ph.D. degrees in management science and engineering from Tsinghua University, Beijing, in 2010. From 2010 to 2012, he was a Research Fellow with the Institute for Media Innovation, Nanyang Technological University, Singapore. From 2012 to 2015, he was a Research Engineer at EON Reality, Singapore. He is currently a Lecturer with Nanjing University of Aeronautics and Astronautics, China. His research interests include virtual reality and human–computer interaction.